Algebraic Sub-structures

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March 2026

1 Group

Definition 1.1. ***(subgroup)***
Let $(G, *)$ *be a group,* $H \subseteq G$*.*
H *is an* ***operational subgroup of G****, if*

1. $H \neq \varnothing$
2. $\forall x, y \in H,\ x^{-1}, xy \in H$

H *is a* ***structural subgroup of G****, if*

1. *There exists map* $\times : H^2 \to H$ *satisfying*
$$h \times h' = h * h', \quad \forall h, h' \in H$$
2. $(H, \times)$ *is a group*

Lemma 1.1. ***(subgroup equivalence)***
Let $(G, *)$ *be a group,* $H \subseteq G$*. Then, T.F.A.E.*

1. H *is an operational subgroup of* G
2. H *is a structural subgroup of* G

If above holds, we denote $H \leq (G, *)$*.*

Proof. [$\Longrightarrow$] Suppose **operational**.
Construct map $\times : H^2 \to H$ as,
$$(h, h') \mapsto h * h'$$
This map is well-defined due to closure under multiplication of H.

W.T.S. $(H, \times)$ is a group.
First, we know $H \neq \varnothing$, so $\exists h_0, h_0^{-1} \in H$, we observe
$$1_G = h_0 * h_0^{-1} \in H$$

1. (associative) $\forall h_1, h_2, h_3 \in H$, we have
$$(h_1 \times h_2) \times h_3 = (h_1 * h_2) * h_3 = h_1 * (h_2 * h_3) = h_1 \times (h_2 \times h_3)$$
2. (identity) $\exists 1_G \in H$, $\forall h \in H$, $h \times 1_G = h * 1_G = h = 1_G * h = 1_G \times h$

3. (inverse) $\forall h \in H$, $\exists h_G^{-1} \in H$, we have

$$h \times h_G^{-1} = h * h_G^{-1} = 1_G = h_G^{-1} * h = h_G^{-1} \times h$$

Thus, $(H, \times)$ is a group. Hence, H is structural.

[$\Longleftarrow$] Suppose **structural**.
Observe we have

$$1_H = 1_H \times 1_H = 1_H * 1_H \implies 1_G = 1_H \in H$$

Since $(H, \times)$ is a group, we know $H \neq \varnothing$.
(closure under multiplication) $\forall a, b \in H$, $a * b = a \times b \in H$.
(closure under inverse) Take any $a \in H$, $\exists a_H^{-1} \in H$ s.t.

$$a * a_H^{-1} = a \times a_H^{-1} = 1_H = 1_G = a_H^{-1} \times a = a_H^{-1} * a \implies a_H^{-1} = a_G^{-1} \in H$$

Hence, H is operational. $\square$

2 Ring

Definition 2.1. ***(subring)***
Let $(R, +, \times)$ be a ring, $S \subseteq R$.
S is an ***operational subring of R****, if*

1. $S \leq (R, +)$
2. $\forall x, y \in S,\ x \times y \in S$

S is a ***structural subring of R****, if*

1. *There exist two maps $\oplus, \otimes : S^2 \to S$ satisfying*

$$s \oplus s' = s + s', \quad s \otimes s' = s \times s', \quad \forall s, s' \in S$$

2. *$(S, \oplus, \otimes)$ is a ring*

Lemma 2.1. ***(subring equivalence)***
Let $(R, +, \times)$ be a ring, $S \subseteq R$. Then, T.F.A.E.

1. *S is an operational subring of R*
2. *S is a structural subring of R*

If above hold, we denote $S \leq_{ring} (R, +, \times)$.

Proof. [$\Longrightarrow$] Suppose **operational**.
Construct map $\oplus, \otimes : S^2 \to S$ as,

$$(s, s') \mapsto_{\oplus} s + s' \quad , \quad (s, s') \mapsto_{\otimes} s \times s'$$

Two maps are well-defined due to closure under addition and multiplication of S.

W.T.S. $(S, \oplus, \otimes)$ is a ring.
First, we know from group section, $(S, \oplus)$ is an abelian additive group.
($\otimes$ associativity) $\forall s_1, s_2, s_3 \in S$, we have

$$(s_1 \otimes s_2) \otimes s_3 = (s_1 \times s_2) \times s_3 = s_1 \times (s_2 \times s_3) = s_1 \otimes (s_2 \otimes s_3)$$

(distributivity) $\forall x, y, z \in S$, we have

$$(x \oplus y) \otimes z = (x + y) \times z = (x \times z) + (y \times z) = (x \otimes z) \oplus (y \otimes z)$$

Similar for the other direction. Thus, S is structural.

[$\Longleftarrow$] Suppose **structural**.
Again by group, we know $S \leq (R, +)$.
Take any $x, y \in S$, observe $x \times y = x \otimes y \in S$.
Thus, S is operational. $\square$

3 Field

Definition 3.1. ***(subfield of a ring)***
Let $(R, +, \times)$ be a ring, $S \subseteq R$.
S is a ***structural subfield of ring R****, if*

1. *There exists two maps $\oplus, \otimes : S^2 \to S$ satisfying*

$$s \oplus s' = s + s', \quad s \otimes s' = s \times s', \quad \forall s, s' \in S$$

2. *$(S, \oplus, \otimes)$ is a field*

Lemma 3.1. ***(basic facts of subfield)***
Let $(F, +, \times)$ be a field, $S \subseteq F$ be a structural subfield of ring F. Then,

1. *$0_F, 1_F \in S$, and $0_S = 0_F$, $1_S = 1_F$*
2. *$(-x)_S = (-x)_F$, $\forall x \in S$*
3. *$x_S^{-1} = x_F^{-1}$, $\forall x \in S - \{0_S\}$*

Proof. Since S is a field, so $0_S, 1_S \in S$ s.t. $0_S \neq 1_S$.
$\implies 0_S \oplus 0_S = 0_S = 0_S + 0_S \implies 0_F = 0_S \in S$
$\implies 1_S \otimes 1_S = 1_S = 1_S \times 1_S \implies 1_F = 1_S \in S$.

Observe $0_F = 0_S = x \oplus (-x)_S = x + (-x)_S \implies (-x)_F = (-x)_S$.
Observe $1_F = 1_S = x \otimes x_S^{-1} = x \times x_S^{-1} \implies x_F^{-1} = x_S^{-1}$. □

Definition 3.2. ***(subfield of a field)***
Let $(F, +, \times)$ be a field, $S \subseteq F$.
S is an ***operational subfield of field F****, if*

1. *$S \leq_{ring} (F, +, \times)$*
2. *$1_F \in S$*
3. *$\forall z \in S - \{0_F\}$, $z_F^{-1} \in S$*

S is a ***structural subfield of field F****, if S is a structural subfield of ring F.*

Lemma 3.2. ***(subfield equivalence)***
Let $(F, +, \times)$ be a field, $S \subseteq F$. Then, T.F.A.E.

1. *S is an operational subfield of field F*
2. *S is a structural subfield of field F*

If above holds, we denote $S \leq_{field} (F, +, \times)$.

Proof. [$\Longrightarrow$] Suppose **operational**.
Then, there exists $\oplus, \otimes : S^2 \to S$ satisfying,

$$(x, y) \mapsto x + y \ \ , \ \ (x, y) \mapsto x \times y$$

And, $(S, \oplus, \otimes)$ is a ring.
($\otimes$ commutativity) obvious.
Let $0_S = 0_F$, $1_S = 1_F$. Note, $0_S \neq 1_S$.
($\otimes$ inverse) obvious.
Then, $(S, \oplus, \otimes)$ is a field. Thus, S is structural.

[$\Longleftarrow$] Suppose **structural**.
Then, $S \leq_{ring} (F, +, \times)$.
By above basic fact lemma, we know $1_F \in S$.
Take any $z \in S - \{0_F\}$, since $0_F = 0_S$, so $\exists z_S^{-1} \in S$; by lemma again, we know $z_S^{-1} = z_F^{-1} \in S$.
Thus, S is operational. $\square$

4 Module

Definition 4.1. ***(submodule)***
Let $(R, M, +_M, \mu)$ be an R-module, $N \subseteq M$.
N is an ***operational R-submodule of M****, if*

1. $N \leq (M, +_M)$
2. $\forall r \in R,\ \forall n \in N,\ \mu(r, n) \in N$

N is a ***structural R-submodule of M****, if*

1. *There exists two maps $+_N : N^2 \to N$, $\nu : R \times N \to N$ satisfying,*

$$n +_N n' = n +_M n', \quad \nu(r, n) = \mu(r, n), \quad \forall r \in R,\ \forall n, n' \in N$$

2. *$(R, N, +_N, \nu)$ is an R-module*

Lemma 4.1. ***(submodule formulation equivalence)***
Let $(R, M, +_M, \mu)$ be an R-module, $N \subseteq M$. Then, T.F.A.E.

1. *N is an operational R-submodule of M*
2. *N is a structural R-submodule of M*

If above holds, we denote $N \leq_{module} (R, M, +_M, \mu)$.

Proof. [$\Longrightarrow$] Suppose **operational**.
Construct map $+_N : N^2 \to N$, $\nu : R \times N \to N$ as,

$$(n, n') \mapsto_{+_N} n +_M n' \quad , \quad (r, n) \mapsto_\nu \mu(r, n)$$

Two maps are well-defined due to closure.
By group, we know $(N, +_N)$ is an abelian group. (abelian check is trivial)
(Distributivity) Take any $r, s \in R$, $n, n' \in N$, observe

$$\nu(r +_R s, n) = \mu(r +_R s, n) = \mu(r, n) +_M \mu(s, n) = \nu(r, n) +_N \nu(s, n)$$

$$\nu(r, n +_N n') = \mu(r, n +_M n') = \mu(r, n) +_M \mu(r, n') = \nu(r, n) +_N \nu(r, n')$$

(action associativity) Take any $r, s \in R$, $n \in N$, observe

$$\nu(r, \nu(s, n)) = \mu(r, \mu(s, n)) = \mu(r \cdot_R s, n) = \nu(r \cdot_R s, n)$$

Suppose R has 1. Observe, $\forall n \in N$,

$$\nu(1_R, n) = \mu(1_R, n) = n$$

Hence, $(R, N, +_N, \nu)$ is an R-module, and N is structural.

[$\Longleftarrow$] Suppose **structural**.
By group, we have $N \leq (M, +_M)$.
Take any $r \in R$, $n \in N$, we have $\nu(r, n) = \mu(r, n) \in N$.
Thus, N is operational. $\square$

5 Algebra

Definition 5.1. ***(D-F R-algebra)***
Let R be a commutative ring with 1_R.
*An **D-F R-algebra** is a ring A with 1_A, with a ring homomorphism (**structure map**) $f : R \to A$ s.t.*

1. $f(1_R) = 1_A$
2. $f(R) \subseteq Z(A) \coloneqq \{z \in A \mid (\forall a \in A)\ za = az\}$

Definition 5.2. ***(A-K R-algebra)***
Let R be a commutative ring with 1_R.
*An **A-K R-algebra** is a D-F R-algebra s.t. its underlying ring is commutative.*

Definition 5.3. ***(Milne F-algebra)***
Let F be a field.
*A **Milne F-algebra** is a ring A with 1_A, s.t.*

1. *F is an operational unital subring of A*
2. $+_F = +_A \restriction F^2$
3. $\cdot_F = \cdot_A \restriction F^2$

Observations
Let R be a commutative ring with 1_R, F be a field.

1. An A-K R-algebra is a D-F R-algebra
2. Let A be a Milne F-algebra. If A is commutative, then $(A, inc : F \to A)$ is an A-K F-algebra

   Proof. Observe $1_A = 1_A \cdot_F 1_F = 1_A \cdot_A 1_F = 1_F$.
   Since F is subring-1 of A, so $inc : F \to A$ is a ring-map-1.
   Thus, we are done □

5.1 Algebra Homomorphism

Definition 5.4. ***(D-F R-algebra homomorphism)***
Let R be a commutative ring with 1_R.
Let (A, f) and (B, g) be D-F R-algebras.
A ***D-F R-algebra homomorphism*** *is a ring homomorphism $\varphi : A \to B$ s.t.*

$$\varphi \circ f = g$$

Definition 5.5. ***(A-K R-algebra homomorphism)***
Let R be a commutative ring with 1_R.
Let (A, f) and (B, g) be A-K R-algebras.
An ***A-K R-algebra homomorphism*** *is a D-F R-algebra homomorphism.*

Definition 5.6. ***(Milne F-algebra homomorphism)***
Let F be a field.
Let A, B be Milne F-algebras.
A ***Milne F-algebra homomorphism*** *is a ring homomorphism $\varphi : A \to B$ s.t.*

$$\varphi(c) = c, \quad \forall c \in F$$

Observations
Let R be a commutative ring with 1_R, F be a field.

1. A D-F R-algebra homomorphism is a ring-map-1 (unital ring homomorphism)

$$1_B = g(1_R) = \varphi(f(1_R)) = \varphi(1_A)$$

2. An A-K R-algebra homomorphism is a D-F R-algebra homomorphism

Lemma 5.1. ***(Milne + commutativity = A-K)***
Let A, B be Milne F-algebras, both commutative, $\varphi : A \to B$ a Milne F-algebra homomorphism.
Then, φ is an A-K F-algebra homomorphism.

Proof. Since A, B are commutative and are Milne F-algebras, F is a subring of both A and B. Let

$$\iota_A : F \hookrightarrow A, \qquad \iota_B : F \hookrightarrow B$$

be the inclusion maps.
Because A is commutative, we have $\iota_A(F) \subseteq Z(A) = A$. Also, since F is a subring of A, the inclusion map ι_A is a ring homomorphism. Hence (A, ι_A) is an A-K F-algebra. Similarly, (B, ι_B) is an A-K F-algebra.
Now let

$$\varphi : A \to B$$

be a Milne F-algebra homomorphism. By definition, φ is a ring homomorphism such that

$$\varphi(c) = c, \qquad \forall c \in F.$$

Therefore, for every $c \in F$,

$$(\varphi \circ \iota_A)(c) = \varphi(\iota_A(c)) = \varphi(c) = c = \iota_B(c).$$

Hence, $\varphi \circ \iota_A = \iota_B$.
So φ is compatible with the structure maps. Therefore φ is an A-K F-algebra homomorphism. □

5.2 Subalgebra

Definition 5.7. ***(D-F operational R-subalgebra)***
Let R be a commutative ring with 1_R.
Let (A, f_A) be a D-F R-algebra, and let $B \subseteq A$.

We say that B is an ***operational D-F*** R***-subalgebra*** *of A iff*

1. *B is an operational unital subring of A;*
2. *$f_A(R) \subseteq B$.*

Definition 5.8. ***(D-F structural R-subalgebra)***
Let R be a commutative ring with 1_R.
Let (A, f_A) be a D-F R-algebra, and let $B \subseteq A$.

We say that B is a ***structural D-F*** R***-subalgebra*** *of A iff*

1. *B is a structural subring of A with $1_A \in B$;*
2. *there exists a map $f_B : R \to B$ such that (B, f_B) is a D-F R-algebra;*
3. *if $\iota : B \hookrightarrow A$ is the inclusion map, then*
$$\iota \circ f_B = f_A.$$

Lemma 5.2. ***(D-F subalgebra bridge lemma)***
Let R be a commutative ring with 1_R. Let (A, f_A) be a D-F R-algebra, and let $B \subseteq A$. Then, T.F.A.E.

1. *B is an operational D-F R-subalgebra of A;*
2. *B is a structural D-F R-subalgebra of A.*

Proof. (1) $\Rightarrow$ (2) Suppose B is an operational D-F R-subalgebra of A.
By definition, B is an operational unital subring of A, and

$$f_A(R) \subseteq B.$$

By the bridge lemma for unital subrings, B is a structural subring of A with $1_A \in B$.
Define a map
$$f_B : R \to B$$
by
$$f_B(r) := f_A(r), \qquad \forall r \in R.$$
This is well-defined because $f_A(R) \subseteq B$.
Since f_A is a ring homomorphism, f_B preserves addition, multiplication, and 1_R. It remains to show that

$$f_B(R) \subseteq Z(B).$$

Take any $r \in R$ and $b \in B$. Since $B \subseteq A$ and $f_A(R) \subseteq Z(A)$, we have

$$f_B(r)b = f_A(r)b = bf_A(r) = bf_B(r).$$

Hence $f_B(r) \in Z(B)$. Therefore
$$f_B(R) \subseteq Z(B),$$
and so (B, f_B) is a D-F R-algebra.

Let

$$\iota : B \hookrightarrow A$$

be the inclusion map. For every $r \in R$, we have

$$(\iota \circ f_B)(r) = \iota(f_B(r)) = \iota(f_A(r)) = f_A(r).$$

Thus

$$\iota \circ f_B = f_A.$$

Therefore B is a structural D-F R-subalgebra of A.

(2) $\Rightarrow$ (1) Suppose B is a structural D-F R-subalgebra of A.
By definition,

1. B is a structural subring of A with $1_A \in B$;
2. there exists a map $f_B : R \to B$ such that (B, f_B) is a D-F R-algebra;
3. if $\iota : B \hookrightarrow A$ is the inclusion map, then
$$\iota \circ f_B = f_A.$$

Since B is a structural subring of A with $1_A \in B$, the bridge lemma for unital subrings implies that B is an operational unital subring of A.
It remains to show that

$$f_A(R) \subseteq B.$$

Take any $r \in R$. Since $f_B(r) \in B$ and ι is the inclusion map, we have

$$f_A(r) = (\iota \circ f_B)(r) = \iota(f_B(r)) = f_B(r) \in B.$$

Hence

$$f_A(R) \subseteq B.$$

Therefore B is an operational D-F R-subalgebra of A. $\square$